TAHIR MAHMOOD

Seoul, Republic of Korea

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Google Scholar | GitHub

PROFESSIONAL EXPERIENCE

Assistant Professor Mar 2023 – Present

Department of Electrical Engineering, Dongguk University, Seoul, South Korea

- Develop course curriculum and teach graduate-level courses.
- Research and development in computer vision and medical image processing.
- Supervise and mentor master's and Ph.D. students, guiding their research work and assisting in academic and professional development.
- Oversee the preparation and submission of research grant proposals to national funding bodies.
- Organize academic seminars and workshops to facilitate knowledge sharing.

Machine learning Engineer Oct 2017 – Feb 2019

Air-fi, Barcelona, Spain

- Assisted in developing and deploying the "AIR-FI Indoor Positioning System".
- Analyzed data and gathered client requirements for system customization.
- Developed a web-based platform for system visualization and management using Python for development and React/Node.js for web-based implementation.

Lecturer Apr 2016 – Sep 2017

Superior University, Lahore, Pakistan

- Taught computer science and software engineering courses.
- Provided guidance and mentorship to students.
- Supervised final-year projects and contributed to the development of industry-relevant curricula.

PUBLICATIONS (FIRST AUTHOR:8, CO-AUTHOR: 14)

First author publications

1. **Tahir Mahmood**, Ganbayar Batchuluun, Seung Gu Kim, Jung Soo Kim, Kang Ryoung Park, "A Lightweight Hierarchical Feature Fusion Network for Surgical Instrument Segmentation in Internet of Medical Things" *Information Fusion* (Impact factor: 15.5)
2. **Tahir Mahmood**, Abdul Wahid, Jin Seong Hong, Seung Gu Kim, Kang Ryoung Park, "A novel convolution transformer-based network for histopathology-image classification using adaptive convolution and dynamic attention", *Engineering Applications of Artificial Intelligence.* [Impact factor: 8]
3. **Tahir Mahmood**, Seung Gu Kim, Se Hyun Nam, and Kang Ryoung Park, "CFFR-Net: A channel-wise features fusion and recalibration network for surgical instruments segmentation", *Engineering Applications of Artificial Intelligence.* [Impact factor: 8]
4. **Tahir Mahmood**, Jiho Choi, and Kang Ryoung Park, "Artificial intelligence-based classification of pollen grains using attention-guided pollen features aggregation network", *Journal of King Saud University – Computer and Information Sciences.* 2023. [Impact factor: 6.9]
5. **Tahir Mahmood**, Se Woon Cho, Kang Ryoung Park, "DSRD-Net: Dual-Stream Residual Dense Network for Semantic Segmentation of Instruments in Robot-Assisted Surgery", *Expert systems with applications*, 2022. [Impact factor: 8.5]

6. **Tahir Mahmood**, Seung Gu Kim, Ja Hyung Koo, and Kang Ryoung Park, "Artificial Intelligence-based Tissue Phenotyping in Colorectal Cancer Histopathology using Visual and Semantic Features Aggregation", *Mathematics*, 2022. [Impact factor: 2.4]
7. **Tahir Mahmood**, Muhammad Owais, Kyoung Jun Noh, Hyo Sik Yoon, Ja Hyung Koo, Adnan Haider, Haseeb Sultan, Kang Ryoun Park, "Accurate Segmentation of Nuclear Regions in Multi-Organ Histopathology Images Using Artificial Intelligence for Cancer Diagnosis in Personalized Medicine", *Journal of Personalized Medicine*, 2021. [Impact factor: 3.4]
8. **Tahir Mahmood**, Muhammad Arsalan, Muhammad Owais, Min Beom Lee, Kang Ryoung Park, "Artificial intelligence-based mitosis detection in breast cancer histopathology images using faster R-CNN and deep CNNs", *Journal of clinical medicine*, 2020. [Impact factor: 3.9]
9. **Tahir Mahmood**, Sheikh Ziauddin, Ahmad R Shahid, Asad Safi, "Mitosis Detection in Breast Cancer Histopathology Images Using Statistical, Color and Shape-Based Features", *Journal of Medical Imaging and Health Informatics*. [Impact factor: 0.659]

Co-author publications (Selected)

1. Abdul Wahid, **Tahir Mahmood**, Jin Seong Hong, Seung Gu Kim, Nadeem Ullah, Rehan Akram, Kang Ryoung Park, "Multi-path residual attention network for cancer diagnosis robust to a small number of training data of microscopic hyperspectral pathological images", Engineering Applications of Artificial Intelligence.
2. Nadeem Ullah, **Tahir Mahmood**, Seung Gu Kim, Se Hyun Nam, Haseeb Sultan, and Kang Ryoung Park, "DCDA-Net: Dual-convolutional dual-attention network for obstructive sleep apnea diagnosis from single-lead electrocardiograms," Engineering Applications of Artificial Intelligence. [Impact factor: 8]
3. Adnan Haider, Muhammad Arsalan, Min Beom Lee, Muhammad Owais, **Tahir Mahmood**, Haseeb Sultan, and Kang Ryoung Park, "Artificial Intelligence-based Computer-aided Diagnosis of Glaucoma Using Retinal Fundus Images, " *Expert systems with applications*, 2022. [Impact factor: 8.665]
4. Muhammad Owais, Hyo Sik Yoon, **Tahir Mahmood**, Adnan Haider, Haseeb Sultan, Kang Ryoung Park, "Light-weighted ensemble network with multilevel activation visualization for robust diagnosis of COVID19 pneumonia from large-scale chest radiographic database", *Applied soft computing* [Impact factor: 8.263]

EDUCATION

Doctor of Engineering Mar 2019 – Feb 2023

Dongguk University, Seoul, South Korea

- Thesis title: A study on deep learning-based classification, detection, and segmentation for histopathology images
- Research contributions: Five first-author, and 12 co-author research papers
- Course work CGPA: 3.9/4.0

Master in Computer Science Sep 2013 – Sep 2015

COMSATS University, Islamabad, Pakistan

- Thesis title: An automated mitosis recognition technique for breast cancer histopathology slide images
- Research contributions: 1 first-author research paper
- Course work CGPA: 3.5/4.0

Bachelor in Computer Science Sep 2009 – Sep 2013

COMSATS University, Islamabad, Pakistan

- Thesis title: Warehouse management system using C#
- Coursework CGPA: 3.1/4.0

Technical Skills

Programming and development skills

- MATLAB (Image processing and computer vision toolbox) [Expert level]
- Python (OpenCV, Numpy, Scikit-learn, TKinter, Pandas, Matplotlib) [Expert level]
- React, Nodejs, Html, JavaScript, PHP [Intermediate level]

Deep learning frameworks

- TensorFlow and Keras (Tensors, data loaders, neural network designing) [Expert level]
- PyTorch (Tensors, data loaders, neural network designing) [Expert level]

Project management: GitHub, GitLab, and Bitbucket

External Collaborators

Dr. Zubair Ahmad: Professor and Consultant Histopathologist at The Aga Khan University Hospital, Pakistan

Dr. Najah Alsubaie: Assistant Professor, Princess Nourah bint Abdulrahman University, Riyadh, Saudi Arabia

Dr. Talat Zehra: Assistant Professor, Jinnah Sindh Medical University, Karachi, Pakistan

Dr. Mahin Shams: Assistant Professor, United Medical and Dental College, Karachi, Pakistan

Dr. Naqqash Dilshad, Assistant Professor, Sejong University, Seoul, Republic of Korea

Participation in Funded Projects

- Deep learning-based semantic classification studies supporting explanatory artificial intelligence and future diagnostic prediction for disease detection in medical images

  Sponsor: National Research Foundation of Korea (NRF-2021R1F1A1045587)
- High-Performance Knowledge System development and human resource training

  Sponsor: Institute for Information & Communications Technology Planning & Evaluation (IITP-2022-2020-0-01789)
- Biometric data generation and spoof detection system evaluation and performance improvement research using GAN (Generative Adversarial Network) and XAI (Explainable AI)

  Sponsor: National Research Foundation of Korea (NRF-2020R1A2C1006179

Awards & Honors

- 2019 Dongguk University SRD (Study and Research at Dongguk) scholarship for Ph.D.
- 2015 Distinction at Department of Computer Science, COMSATS University, Islamabad, Pakistan.
- 2009 National ICT R&D scholarship for Bachelor in COMSATS University, Islamabad, Pakistan.

Languages

English: *Proficient,* Spanish: *Intermediate,* Korean: *Intermediate*

References

Dr. Muhammad Arsalan (PhD Mentor)
Research Assistant Professor,
KINDI Center for Computing Research, College of Engineering, Qatar University, Qatar
Email: muhammad.arsalan@qu.edu.qa

Dr. Muhammad Owais (PhD Mentor)
Postdoctoral Fellow,
Khalifa University, UAE
Email: muhammad.owais@ku.ac.ae

No. 6311

This Diploma makes known that

Dongguk University

on the recommendation of the faculty has conferred upon

TAHIR MAHMOOD

the degree of

Doctor of Engineering

and the honors, rights and privileges of the degree given by Dongguk University, Korea
in witness whereof the signatures of the President and the Dean of the Graduate School are herewith affixed
this sixteenth day of February in the year of two thousand and twenty-three

Han Hee Won
Dean of the Graduate School

dongguk UNIVERSITY

President

Serial No: 034203

Registration No: CIIT/FA13-RCS-027/ISB

By the Authority of the Board of Governors

COMSATS Institute of Information Technology

TAHIR MAHMOOD s/o GUL RAHMAN

Of

Islamabad Campus

has been conferred upon the degree of

Master of Science in Computer Science

Given on this Fifteenth day of September two thousand and Fifteen at Islamabad

Date of Issuance: 15th September 2015

Controller of Examinations

Rector

Registrar

Serial No: 024701

Registration No: CIIT/FA09-BCS-178/ISB

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